Software Requirements Specification

Help Desk System

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Document Information

Field	Value
Document Title	Software Requirements Specification (SRS) — Help Desk System
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System	Help Desk System
Purpose	Define the functional and non-functional requirements for a system used to submit, track, assign, and manage support tickets.
Revision Summary	Restricted ticket assignment to Administrator only. Removed "Authorized Team Lead" role. Updated roles to three (Requester, Support Agent, Administrator). Added audit fields for assignment tracking.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the requirements of the Help Desk System. It defines the expected behavior, user roles, functional capabilities, acceptance criteria, and quality requirements. The document is intended for stakeholders, analysts, designers, developers, and testers.

1.2 Scope

The Help Desk System enables employees to report technical issues through support tickets and follow their progress. Support agents can review assigned tickets, communicate with requesters, update ticket status and priority, and receive notifications. Administrators manage user accounts, control access to the system, and are the only role authorized to assign tickets to support agents.

1.3 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
User Story	A short description of a capability from the perspective of a user role
Acceptance Criteria	Conditions that must be satisfied for a user story to be considered complete
Ticket	A record representing a reported issue or support request
Requester	A user who submits a support ticket
Support Agent	A user responsible for handling assigned support tickets
Administrator	A user authorized to manage users, roles, and ticket assignments
Assignment	The act of linking a ticket to a specific support agent, performed only by an Administrator
Audit Trail	A record of important ticket changes including the acting user and timestamp

2. Overall Description

2.1 Product Perspective

The Help Desk System is a centralized application for managing support requests. It provides a shared workflow for requesters, support agents, and administrators.

2.2 Product Functions

- Authenticate users and manage their profiles.
- Create, view, search, filter, and update support tickets.
- Attach files to tickets and exchange comments.
- Assign tickets to support agents (Administrator only) and update ticket status and priority.
- Notify users about relevant ticket events.
- Allow administrators to manage user accounts and roles.

2.3 Operating Environment

The system is expected to be accessible through a modern web browser on desktop or mobile devices. A server-side application and database are required to store user, ticket, comment, attachment metadata, notification, and audit data.

3. User Roles

The system defines **three** user roles. There is no separate "Team Lead" role; all assignment privileges belong exclusively to the Administrator.

Role	Responsibilities
Requester / Employee REQ	Creates support tickets, attaches files, views and updates eligible tickets, adds comments, and receives notifications.
Support Agent AGENT	Views assigned tickets, adds comments, changes status and priority, communicates with requesters, and receives notifications. Cannot assign tickets.
Administrator ADMIN	Manages user accounts and roles, activates/deactivates users, assigns tickets to support agents, and performs all other authorized administrative actions. Has all Support Agent privileges.

4. Functional Requirements

4.1 Authentication and Profile Management

- **FR-01:** The system shall allow registered users to log in with valid credentials.
- **FR-02:** The system shall allow authenticated users to log out securely.
- **FR-03:** The system shall allow users to view and update their permitted profile information.

4.2 Ticket Management

- **FR-04:** The system shall allow employees to create support tickets with a title, description, and category.
- **FR-05:** The system shall support file attachments subject to configured type and size limits.
- **FR-06:** The system shall allow requesters to view their own tickets.
- **FR-07:** The system shall provide search and filtering by ticket number/title, status, and priority, subject to access permissions.
- **FR-08:** The system shall allow requesters to update permitted ticket details while the ticket is editable.

4.3 Agent Operations

- **FR-09:** The system shall allow **Administrators only** to assign tickets to **active** support agents.
- **FR-10:** The system shall allow agents to view tickets assigned to them.
- **FR-11:** The system shall allow authorized agents to update ticket status within the configured workflow.
- **FR-12:** The system shall allow authorized agents to set ticket priority to Low, Medium, High, or Critical.

4.4 Communication and Administration

- **FR-13:** The system shall allow authorized participants to add comments to tickets and shall record the author and timestamp.
- **FR-14:** The system shall create and display ticket notifications for relevant users.
- **FR-15:** The system shall allow administrators to view, create, update, activate, and deactivate user accounts and assign roles.

5. User Stories and Acceptance Criteria

Each user story below includes acceptance criteria that can be used to verify whether the requirement has been implemented correctly. Development task lists are included as planning guidance and may be converted into Trello checklist items.

US-01 — User Login

Category: Authentication

User Story: As a user, I want to log in to the system so that I can access my account and support features.

Acceptance Criteria

AC-1 Valid credentials allow the user to log in.

AC-2 Invalid credentials display an appropriate error message.

AC-3 Unauthenticated users cannot access protected pages.

AC-4 After login, the user is redirected according to their role.

Development Tasks (Planning Checklist)

- Design the Login UI
- Implement the Login API
- Validate user credentials
- Implement authentication
- Write unit tests
- Test the acceptance criteria

US-02 — User Logout

Category: Authentication

User Story: As a logged-in user, I want to log out securely so that unauthorized people cannot access my account.

Acceptance Criteria

AC-1 Selecting Logout terminates the active session or invalidates the authentication token.

AC-2 The user is redirected to the Login page.

AC-3 The user cannot access protected pages using the terminated session.

AC-4 The system displays a logout confirmation message.

Development Tasks (Planning Checklist)

- Add a Logout button to the interface
- Implement the Logout endpoint/action
- Terminate the session or invalidate the token
- Redirect to Login
- Test logout and protected-page access

US-03 — View and Update Profile

Category: Authentication

User Story: As a user, I want to view and update my profile so that my personal information remains accurate.

Acceptance Criteria

AC-1 The user can view their name, email, and other permitted profile details.

AC-2 The user can update editable profile fields.

AC-3 The system validates required fields and email format.

AC-4 Valid changes are saved and a success message is displayed.

AC-5 Users cannot modify another user's profile without authorization.

Development Tasks (Planning Checklist)

- Design the Profile page
- Implement the Get Profile API
- Implement the Update Profile API
- Validate profile data and email
- Test profile viewing and updating
- Test unauthorized access

US-04 — Create a Support Ticket

Category: Ticket Management

User Story: As an employee, I want to create a support ticket so that I can report a technical issue.

Acceptance Criteria

AC-1 The requester can provide a title, description, and category.

AC-2 The system validates required fields.

AC-3 The system generates a unique ticket number.

AC-4 A newly created ticket has the initial status **Open**.

AC-5 A newly created ticket has **no assigned agent**.

AC-6 The system displays a success message when creation succeeds.

AC-7 If creation fails, an error is shown and no incomplete ticket is saved.

Development Tasks (Planning Checklist)

- Design the Create Ticket form
- Create the Ticket entity and database table
- Implement the Create Ticket API
- Validate required fields
- Generate a unique ticket number
- Set initial status to Open
- Write unit/integration tests
- Test successful and failed submissions

US-05 — Attach Files to a Ticket

Category: Ticket Management

User Story: As a requester, I want to attach screenshots or documents to a ticket so that the support agent can understand my problem better.

Acceptance Criteria

AC-1 The system accepts only configured supported file types.

AC-2 The system rejects files exceeding the configured size limit.

AC-3 Unsupported file types are rejected with an error message.

AC-4 Uploaded files are associated with the correct ticket.

AC-5 The system displays an error if an upload fails.

AC-6 Only authorized users can access ticket attachments.

Development Tasks (Planning Checklist)

- Add a file upload control

- Define supported file types and maximum size
- Implement the upload API
- Validate file type and size on the server
- Store files securely
- Associate attachments with the ticket
- Implement attachment access authorization
- Test valid and invalid files and upload failures

US-06 — View My Tickets

Category: Ticket Management

User Story: As a requester, I want to view my submitted tickets so that I can monitor their progress.

Acceptance Criteria

AC-1 The system displays tickets created by the logged-in requester.

AC-2 Each ticket displays its number, title, creation date, priority, and status.

AC-3 Users cannot access other users' private tickets unless authorized.

AC-4 The requester can open a ticket to view its details.

AC-5 An informative empty-state message appears when the requester has no tickets.

Development Tasks (Planning Checklist)

- Design the My Tickets page
- Implement the Get My Tickets API
- Retrieve tickets for the current requester
- Display ticket fields
- Implement ticket-details navigation
- Handle the empty state
- Test authorization and ticket visibility

US-07 — Search and Filter Tickets

Category: Ticket Management

User Story: As a user, I want to search and filter tickets so that I can find a specific support request quickly.

Acceptance Criteria

AC-1 The user can search by ticket number or title.

AC-2 The user can filter tickets by status and priority.

AC-3 Search results contain only tickets the user is authorized to view.

AC-4 Results reflect the active search term and selected filters.

AC-5 The system displays a message when no tickets match.

AC-6 The user can clear filters and restore the default results.

Development Tasks (Planning Checklist)

- Add a search field
- Add status and priority filters
- Implement search and filter logic
- Update the API to support search and filters
- Enforce ticket permissions
- Add a Clear Filters action
- Test combined filters and empty results

US-08 — Update Ticket Details

Category: Ticket Management

User Story: As a requester, I want to update my ticket details when necessary so that I can provide additional or corrected information.

Acceptance Criteria

AC-1 The requester can edit permitted fields while the ticket is in an editable state.

AC-2 The system validates updated data.

AC-3 Users cannot modify tickets they do not own unless authorized.

AC-4 Valid changes are saved and a success message is displayed.

AC-5 Editing is prevented when the ticket is in a non-editable state.

AC-6 The system displays an error message if the update fails.

Development Tasks (Planning Checklist)

- Define editable ticket fields
- Design the Edit Ticket form
- Implement the Update Ticket API
- Validate updated data
- Enforce ownership and edit permissions

- Restrict editing by ticket status
- Display success/error messages
- Test valid updates and unauthorized access

US-09 — Assign Tickets to Support Agents REVISED

Category: Agent Operations

User Story: As an **Administrator**, I want to assign tickets to support agents so that each issue has a responsible person.

Acceptance Criteria

AC-1 **Only an Administrator** can assign a ticket to an active support agent.

AC-2 The assigned agent is recorded against the ticket.

AC-3 The system records **who performed the assignment** and **when** it occurred.

AC-4 The assigned agent receives a notification.

AC-5 The system prevents assignment to inactive agents.

AC-6 The system prevents assignment to users who do not have the Support Agent role.

AC-7 The system prevents assignment by any user who is not an Administrator.

AC-8 The updated assignment is displayed in ticket details.

AC-9 The system displays a success or error message after the assignment attempt.

Development Tasks (Planning Checklist)

- Design the Assign Agent interface (Admin only)
- Implement the assignment API with `[Authorize(Roles = "Administrator")]`
- Retrieve active support agents only
- Validate agent status and role
- Save the assigned agent, assigner, and timestamp
- Notify the assigned agent
- Record the assignment in the audit trail
- Test successful and rejected assignments
- Test unauthorized access attempts

US-10 — View Assigned Tickets

Category: Agent Operations

User Story: As a support agent, I want to view my assigned tickets so that I can manage the issues I am responsible for.

Acceptance Criteria

AC-1 The system displays tickets assigned to the logged-in agent.

AC-2 Each ticket displays its title, requester, priority, status, and creation date.

AC-3 The agent can open a ticket to view its details.

AC-4 Restricted administrative information is not shown to unauthorized agents.

AC-5 An empty-state message appears when no tickets are assigned.

AC-6 Agents cannot access other agents' restricted tickets unless authorized.

Development Tasks (Planning Checklist)

- Design the Assigned Tickets page
- Implement the Get Assigned Tickets API
- Retrieve tickets assigned to the current agent
- Display ticket details
- Implement details navigation
- Handle the empty state
- Enforce authorization
- Test access to other agents' tickets

US-11 — Change Ticket Status

Category: Agent Operations

User Story: As a support agent, I want to update a ticket's status so that the requester can track its progress.

Acceptance Criteria

AC-1 Authorized agents can change status according to the defined workflow.

AC-2 The supported statuses are Open, In Progress, Pending, Resolved, and Closed.

AC-3 The system records the new status, timestamp, and user who made the change.

AC-4 The requester is notified of significant status changes.

AC-5 The system rejects invalid status transitions.

AC-6 Only authorized users can change ticket status.

AC-7 The system displays success or error feedback.

Development Tasks (Planning Checklist)

- Define valid status transitions
- Add a status selector
- Implement the Update Ticket Status API
- Validate transitions on the server
- Record status history and timestamps
- Trigger notifications
- Enforce authorization
- Write unit/integration tests

US-12 — Set Ticket Priority

Category: Agent Operations

User Story: As a support agent, I want to set ticket priority so that urgent issues can be handled first.

Acceptance Criteria

AC-1 The supported priority levels are Low, Medium, High, and Critical.

AC-2 Authorized support agents can change ticket priority.

AC-3 The updated priority is saved and displayed.

AC-4 Unauthorized users cannot change priority.

AC-5 The system displays a success message after a valid update.

AC-6 The system displays an error message if the update fails or the value is invalid.

Development Tasks (Planning Checklist)

- Define priority levels
- Add a priority selector
- Implement the Update Ticket Priority API
- Validate priority values on the server
- Save and display the updated priority
- Enforce authorization
- Test all priority levels and invalid values

US-13 — Add Comments to a Ticket

Category: Communication

User Story: As a requester or support agent, I want to add comments to a ticket so that both parties can exchange information.

Acceptance Criteria

AC-1 Authorized participants can add comments to a ticket.

AC-2 The system stores each comment with its author and timestamp.

AC-3 Comments appear in chronological order.

AC-4 The system validates that a comment is not empty.

AC-5 Internal notes are visible only to authorized support staff and administrators.

Development Tasks (Planning Checklist)

1. Design the comments section in the ticket details page
2. Implement the Add Comment API
3. Store each comment with its author and timestamp
4. Display comments in chronological order
5. Validate that comments are not empty
6. Implement internal-notes visibility and authorization
7. Test comment creation and access permissions

US-14 — Receive Ticket Notifications

Category: Communication

User Story: As a requester or support agent, I want to receive notifications about ticket updates so that I can stay informed about important changes.

Acceptance Criteria

AC-1 The requester is notified when the ticket status changes.

AC-2 The assigned support agent receives a notification when a ticket is assigned to them.

AC-3 Notifications include the ticket number and a brief description of the update.

AC-4 Users can view their notifications in the system.

AC-5 Users cannot view notifications belonging to other users.

AC-6 The system displays an appropriate message if notifications cannot be delivered.

Development Tasks (Planning Checklist)

- Design the notifications interface

- Create the Notification entity and database table
- Implement notification creation logic
- Generate notifications when ticket status changes
- Notify agents when tickets are assigned to them
- Implement the Get My Notifications API
- Enforce notification access permissions
- Handle notification delivery failures
- Test notification creation and authorization

US-15 — Manage User Accounts

Category: Administration

User Story: As an administrator, I want to manage user accounts so that I can control access to the Help Desk System.

Acceptance Criteria

AC-1 Administrators can view the list of registered users.

AC-2 Administrators can create new user accounts with valid information.

AC-3 Administrators can update user details and assign appropriate roles.

AC-4 Administrators can activate or deactivate user accounts.

AC-5 The system validates required fields and email formats.

AC-6 Only authorized administrators can manage user accounts.

AC-7 The system displays a success or error message after each operation.

Development Tasks (Planning Checklist)

- Design the User Management interface
- Implement the Get All Users API
- Implement the Create User API
- Implement the Update User API
- Implement account activation and deactivation
- Implement role assignment
- Validate user information and email format
- Enforce administrator authorization
- Write unit and integration tests

6. Non-Functional Requirements

ID	Requirement
NFR-01	**Security:** The system shall require authentication for protected features and enforce role-based and resource-level authorization.
NFR-02	**Data Validation:** The system shall validate user input on the server and return clear validation errors without exposing sensitive implementation details.
NFR-03	**Performance:** For typical usage and normal network conditions, common pages and ticket searches should respond within an agreed target time defined during design and testing.
NFR-04	**Availability and Reliability:** The system should handle expected operational errors gracefully and avoid saving incomplete or inconsistent records.
NFR-05	**Usability:** The user interface should provide clear labels, understandable error messages, and feedback after successful or failed actions.
NFR-06	**Privacy:** Users shall only access profile data, tickets, comments, attachments, and notifications permitted by their role and ownership.
NFR-07	**Maintainability:** The application should be organized into maintainable components and include automated tests for important business rules.
NFR-08	**Compatibility:** The web interface should work with current major browsers and adapt to desktop and mobile screen sizes.
NFR-09	**Auditability:** Important ticket changes — including status updates, priority changes, and assignments — shall record the acting user and timestamp.
NFR-10	**File Safety:** Uploaded files should be validated and stored in a way that prevents unauthorized access and reduces file-upload risks.

7. Assumptions and Constraints

- Users have access to a supported web browser and network connection.
- Each user has an account and a role that determines available features.
- Ticket categories are configured by an authorized administrator or system configuration.
- Supported attachment types and maximum file size will be defined during implementation.
- Email, SMS, or external push notifications are not assumed unless a delivery provider is selected; in-system notifications are included in scope.

- Ticket status transitions and which roles may perform them must be finalized before implementation.
- **Ticket assignment is restricted to Administrators only.** No other role may assign tickets to support agents.
- Personal data and ticket attachments must be protected according to the organization's privacy and security policies.

8. Future Enhancements

- Introduce a **Team Lead** role with delegation privileges for ticket assignment.
- Manage ticket categories from the administration panel.
- Generate support reports and dashboards.
- Provide a searchable knowledge base and frequently asked questions.
- Configure service-level agreements (SLAs) and escalation rules.
- Integrate email notifications or other external communication channels.
- Allow requesters to select a preferred agent during ticket creation (subject to administrator approval).

Appendix A — Suggested Ticket Status Workflow

```
Open  →  In Progress  →  Pending / Resolved  →  Closed
```

This is a high-level suggested workflow. The permitted transitions (including reopening a ticket) should be explicitly agreed upon and enforced by the application.

Proposed Valid Transitions

From	To	Allowed Roles
Open	In Progress	Support Agent, Administrator
In Progress	Pending	Support Agent, Administrator
In Progress	Resolved	Support Agent, Administrator
Pending	In Progress	Support Agent, Administrator
Pending	Resolved	Support Agent, Administrator
Resolved	Closed	Support Agent, Administrator
Resolved	Open (Reopen)	Administrator only
Closed	Open (Reopen)	Administrator only

Appendix B — Business Rules for Assignment

ID	Rule
BR-01	Only an **Administrator** can assign tickets to a support agent.
BR-02	A ticket cannot be assigned to a user whose account is **inactive** (`IsActive = false`).
BR-03	A ticket cannot be assigned to a user who does not have the **Support Agent** role.
BR-04	The assigned agent must receive a **notification** after the assignment.
BR-05	The system must record the **assigner**, the **assignee**, and the **timestamp** of the assignment.
BR-06	A ticket may be **reassigned** from one agent to another; the previous assignment is preserved in the audit trail.
BR-07	An assignment may be **removed** (unassigned) by an Administrator; this action is also recorded in the audit trail.

Appendix C — Permission Matrix

Capability	Requester	Support Agent	Administrator
Login / Logout	✅	✅	✅
View / Update Own Profile	✅	✅	✅
Create Ticket	✅	❌	❌
View Own Tickets	✅	❌	❌
View Assigned Tickets	❌	✅	✅
View All Tickets	❌	❌	✅
Update Own Ticket Details	✅	❌	❌
Upload Attachments	✅	❌	❌
Assign Ticket to Agent	❌	❌	✅
Change Ticket Status	❌	✅	✅
Set Ticket Priority	❌	✅	✅
Add Comments	✅	✅	✅
Add Internal Notes	❌	✅	✅
Receive Notifications	✅	✅	✅
Manage Users (CRUD)	❌	❌	✅
Activate / Deactivate Users	❌	❌	✅
Assign Roles	❌	❌	✅

Appendix D — Data Model (ERD Summary)

The system uses ASP.NET Core Identity for authentication and authorization, extended with custom domain entities for ticket management. The tables below summarize the main entities and their relationships.

Identity Entities

Entity	Key Attributes	Relationships
ApplicationUser	Id, FullName, Email, PasswordHash, IsActive, CreatedAt, UpdatedAt, RoleId	HasRole → Role; Creates → Ticket; Handles → Ticket; Writes → Comment; Uploads → Attachment; Receives → Notification; PerformedChanges → TicketStatusHistory
ApplicationRole	Id, Name, Description, CreatedAt	AssignedToUsers → User

Domain Entities

Entity	Key Attributes	Relationships
Ticket	TicketId, TicketNumber, Title, Description, CategoryId, RequesterId, AssignedAgentId, AssignedBy, AssignedAt, StatusId, Priority, CreatedAt, UpdatedAt	CreatedBy → User; AssignedTo → User; BelongsToCategory → Category; HasStatus → TicketStatus; HasComments → Comment; HasAttachments → Attachment; TriggersNotification → Notification; HasStatusHistory → TicketStatusHistory
Category	CategoryId, CategoryName, IsActive	ContainsTickets → Ticket
TicketStatus	StatusId, StatusName (Open, In Progress, Pending, Resolved, Closed)	AppliedToTickets → Ticket
Attachment	AttachmentId, TicketId, FileName, FilePath, FileType, FileSize, UploadedBy, UploadedAt	AttachedTo → Ticket; UploadedBy → User
Comment	CommentId, TicketId, AuthorId, CommentText, IsInternal, CreatedAt	CommentedOn → Ticket; WrittenBy → User
Notification	NotificationId, UserId, TicketId, Message, IsRead, CreatedAt	NotifiesAbout → Ticket; ReceivedBy → User
TicketStatusHistory	HistoryId, TicketId, Action, OldStatusId, NewStatusId, ChangedBy, ChangedAt, Notes	HistoryOf → Ticket; ChangedFrom → TicketStatus; ChangedTo → TicketStatus; ChangedBy → User

Key Relationship Names

Relationship	Entities	Cardinality
HasRole	User → Role	M : 1
Creates	User (Requester) → Ticket	1 : M
AssignedTo	Ticket → User (Agent)	M : 1
BelongsToCategory	Ticket → Category	M : 1
HasStatus	Ticket → TicketStatus	M : 1
HasComments	Ticket → Comment	1 : M
HasAttachments	Ticket → Attachment	1 : M
TriggersNotification	Ticket → Notification	1 : M
HasStatusHistory	Ticket → TicketStatusHistory	1 : M
ReceivedBy	Notification → User	M : 1
ChangedBy	TicketStatusHistory → User	M : 1

> **Note:** The `TicketStatusHistory` entity includes an `Action` field to record the type of change — *StatusChanged*, *PriorityChanged*, *Assigned*, *Unassigned*, *CommentAdded*, or *AttachmentAdded*. This supports the auditability requirement (NFR-09).

> **End of Document** — Help Desk System SRS v2.0